

Week 10. R Practice

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```
library(dplyr)
library(ggplot2)
library(DescTools)
# install.packages("SuppDists")
library(SuppDists)
# install.packages("combinat")
library(combinat)
# install.packages("asht")
library(asht)
```

Paired comparison

Suppose twins were involved in a study of cholesterol-reducing drugs. One of the twins in each pair was given drug 1 X_i and the other was given drug 2 Y_i , where the choice was made at random.

```
twins <- read.csv("twins.csv")
head(twins)
```

	X	Y
1	74	63
2	55	58
3	61	49
4	41	47
5	53	50
6	74	69

$D_i = Y_i - X_i$ 들이 서로 독립이며 중앙값 θ 에 대해 대칭인 분포 F_i 를 따를 때 $H_0: \theta = 0$ 를 검정하고자 한다.

```
x <- twins$X; y <- twins$Y
```

(1) Paired t-test ; D_i 의 평균이 0인지를 검정

```
t.test(y, x, paired=T) # t.test(y-x)
```

$H_0: E D_i = 0$ vs. $H_1: E D_i \neq 0$

Paired t-test

data: y and x

t = -2.5475, df = 16, p-value = 0.02151

두 약의 효과에 차이가 있다?

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

-12.178505 -1.115612

sample estimates:

$n=17$ 로 작은데 정규성 가정은 어떨까?
 $\Rightarrow$ 순위기반 검정도 같이보자.

mean difference
-6.647059

(2) Wilcoxon signed-rank test (가정 D_i 의 분포는 symmetric (대칭))

```
Ri <- rank(abs(y-x))
Vi <- sign(y-x)
T.SR <- sum(Ri[Vi>0])

Z.SR <- (T.SR-sum(Ri)/2)/sqrt(sum(Ri^2)/4)
( pnorm(abs(Z.SR), lower.tail=F)*2 )
```

[1] 0.02758953 : 대칭분 검사

```
wilcox.test(y, x, paired=TRUE)$p.value # large sample test
```

Warning in wilcox.test.default(y, x, paired = TRUE): cannot compute exact
p-value with ties

[1] 0.02930307 : R에서의 계산

```
asht::wsrTest(y-x)$p.value # exact P-value
```

[1] 0.02563477 : exact p-value

전부 0.05보다 작으니 두 약의 효과에 차이가 있다.

(3) Sign test (Binomial test) 대칭성 가정 필요 X

```
n <- length(x)
SN <- sum(Vi[Vi>0])

binom.test(SN, n=n, p=0.5)$p.value # exact P-value
```

[1] 0.1434631 > 0.05

```
Z.SN <- (SN-n/2)/sqrt(n/4)
( pnorm(abs(Z.SN), lower.tail=F)*2 ) # approximated P-value
```

[1] 0.08955507 > 0.05 H₀ 기각 불가 in sign test

Note: Sign test does not require a symmetric distribution for D_i .

Randomized Complete Block Design (RCBD)

In twins example, there are 2 treatments and 17 blocks (pairs).

Extension: Matched Pairs Design $\Rightarrow$ RCBD

The data in "hay.csv" are dry matter contents (in kilograms) of hay obtained from experimental plots. The experiment was designed as a randomized complete block, and the treatments are three cutting dates (Sept.1, Sept.15, Sept.30).

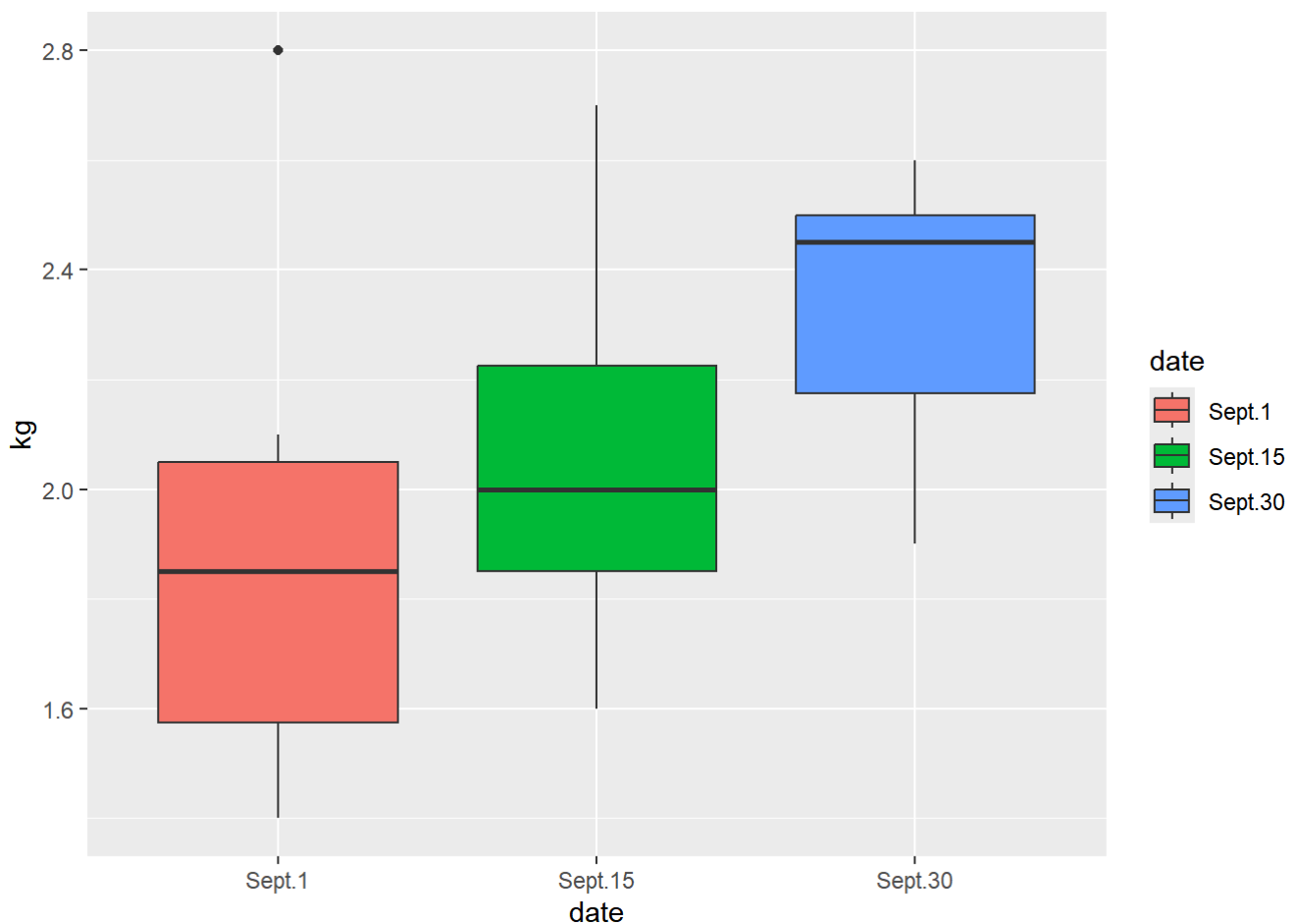
```
hay <- read.csv("hay.csv")
head(hay)
```

block treatment

	kg	plot	date
1	1.5	1	Sept.1
2	2.1	2	Sept.1
3	1.9	3	Sept.1
4	2.8	4	Sept.1
5	1.4	5	Sept.1
6	1.8	6	Sept.1

```
hay <- hay %>% mutate(across(.cols=c(2:3), .fns=factor))

ggplot(hay, aes(x=date, y=kg, fill=date)) + geom_boxplot()
```



관측값들에 대해 다음 모형을 가정하자.

$$Y_{ij} = \mu + t_j + b_j + \epsilon_{ij}$$

- μ is an overall effect
- t_i 's are the treatment effects
- b_k 's are the block effects

- ϵ_{ij} 's are i.i.d with median 0

$H_0 : t_1 = \dots = t_3$ 를 검정하기 위해 다음의 test들을 고려할 수 있다.

(1) F-test

```
## Define a function to compute F-statistic
getF.CBD <- function(x, treatment, block){
```

```
  mean.trt <- tapply(x, treatment, mean)
  mean.block <- tapply(x, block, mean)
  mean.overall <- mean(x)
  k <- length(mean.trt); b <- length(mean.block)
  SSTreatment <- b*sum( (mean.trt - mean.overall)^2 )
  SSBlock = k * sum( (mean.block - mean.overall )^2 )
  SST = sum( (x - mean.overall)^2 )
  SSE = SST - SSTreatment - SSBlock
  return( (SSTreatment/(k-1)) / (SSE/((k-1)*(b-1))) )
}
```

$$SST = SST_{\text{treatment}} + SS_{\text{Block}} + SSE$$

$$F = \frac{SST_{\text{treatment}} / (k-1)}{SSE / ((k-1)(b-1))}$$

k : treatment 개수 (= 3)
b : block 개수 (= Plot 개수)

```
# classical F-test
k <- length(unique(hay$date)); b <- length(unique(hay$plot))
F.obs <- getF.CBD(hay$kg, hay$date, hay$plot)
( p.value <- 1-pf(F.obs, k-1, (k-1)*(b-1)) )
```

[1] 0.01292694 < 0.05 ~ treatment의 효과가 유의하게 존재.

```
# permutational F-test
Num.perm <- 1999
F.perm <- vector(length=Num.perm)
for (i in 1:Num.perm) {
```

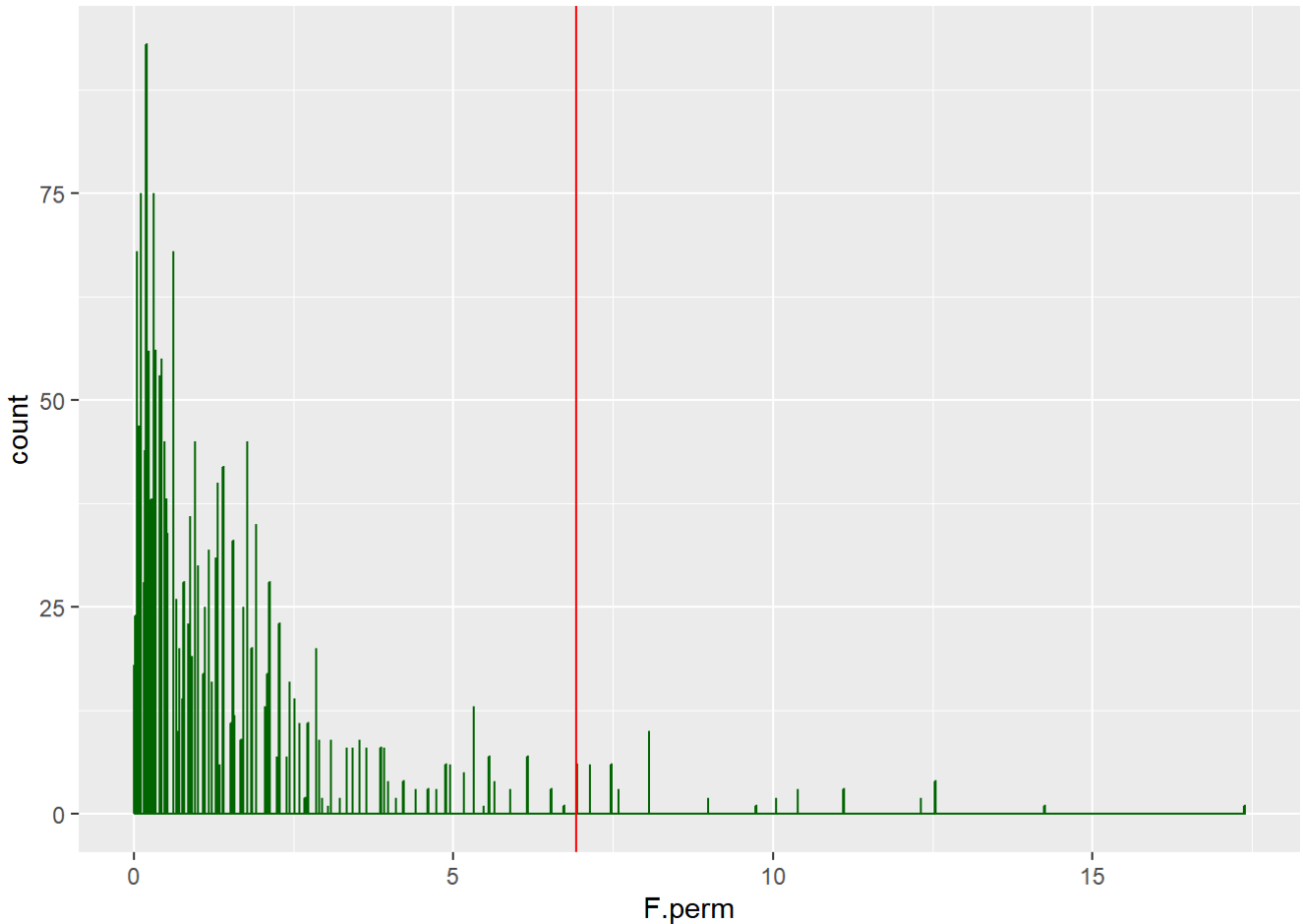
but, F-test는 등분산성, 정규성 가정 의존
~ permutation test도 같이 보자.

```
  plots <- unique(hay$plot)
  date.perm <- hay$date

  # block-wise randomization
  for (j in 1:length(plots)) {
    date.perm[hay$plot == plots[j]] <- sample(hay$date[hay$plot == plots[j]], replace=FALSE)
  }
```

```
  F.perm[i] <- getF.CBD(hay$kg, date.perm, hay$plot)
}
```

```
ggplot(data.frame(F.perm), aes(F.perm)) +
  geom_histogram(binwidth = 0.005, color="darkgreen") +
  geom_vline(xintercept=F.obs, color='red')
```



```
( p.value <- (sum(F.perm>=F.obs)+1)/(Num.perm+1) )
```

[1] 0.0225 $< 0.05 \rightarrow \text{reject } H_0 \Rightarrow \text{treatment 효과 유의하게 존재.}$

(2) Friedman test

```
friedman.test(kg ~ date | plot, data=hay)
```

Friedman rank sum test

data: kg and date and plot

Friedman chi-squared = 4.3333, df = 2, p-value = 0.1146

F -test: 차이의 크기를 사용
Friedman: 차이의 순위만 사용.
(크기 무시)

70.05 Friedman test 기준 기준값

```
# friedman.test(y=hay$kg, groups=hay$date, blocks=hay$plot)
```

treatment (cutting date)의

(3) Aligned rank test ; block 평균을 빼고 전체 값들의 순위를 매김

효과가 유의하지 않다.

```
## Define a function to compute Q-statistic
```

```
getAlignedQ.CBD <- function(x, treatment, block){
```

```
  k <- length(unique(treatment)); b <- length(unique(block))
```

```
  df <- data.frame(x=x, treatment=treatment, block=block)
```

why? 차이의 크기 정보가
중요한데는 Friedman
test가 차이에 대해
덜 민감하게 판단.

```

# rank blockwise-mean adjusted obs.
df <- df %>% group_by(block) %>% mutate(x.adjusted = x-mean(x))
df$Rij <- rank(df$x.adjusted)
df <- df %>% group_by(treatment) %>% mutate(Ri = mean(Rij))
df <- df %>% group_by(block) %>% mutate(Rj = mean(Rij))

Q <- ( (k-1)/sum((df$Rij-df$Rj)^2) ) * sum(b^2*(df$Ri-(b*k+1)/2)^2)

return(Q)
}

( Q.obs <- getAlignedQ.CBD(hay$kg, hay$date, hay$plot) )

```

[1] 42.24083

```

Num.perm <- 1999
Q.perm <- vector(length=Num.perm)
for (r in 1:Num.perm) {

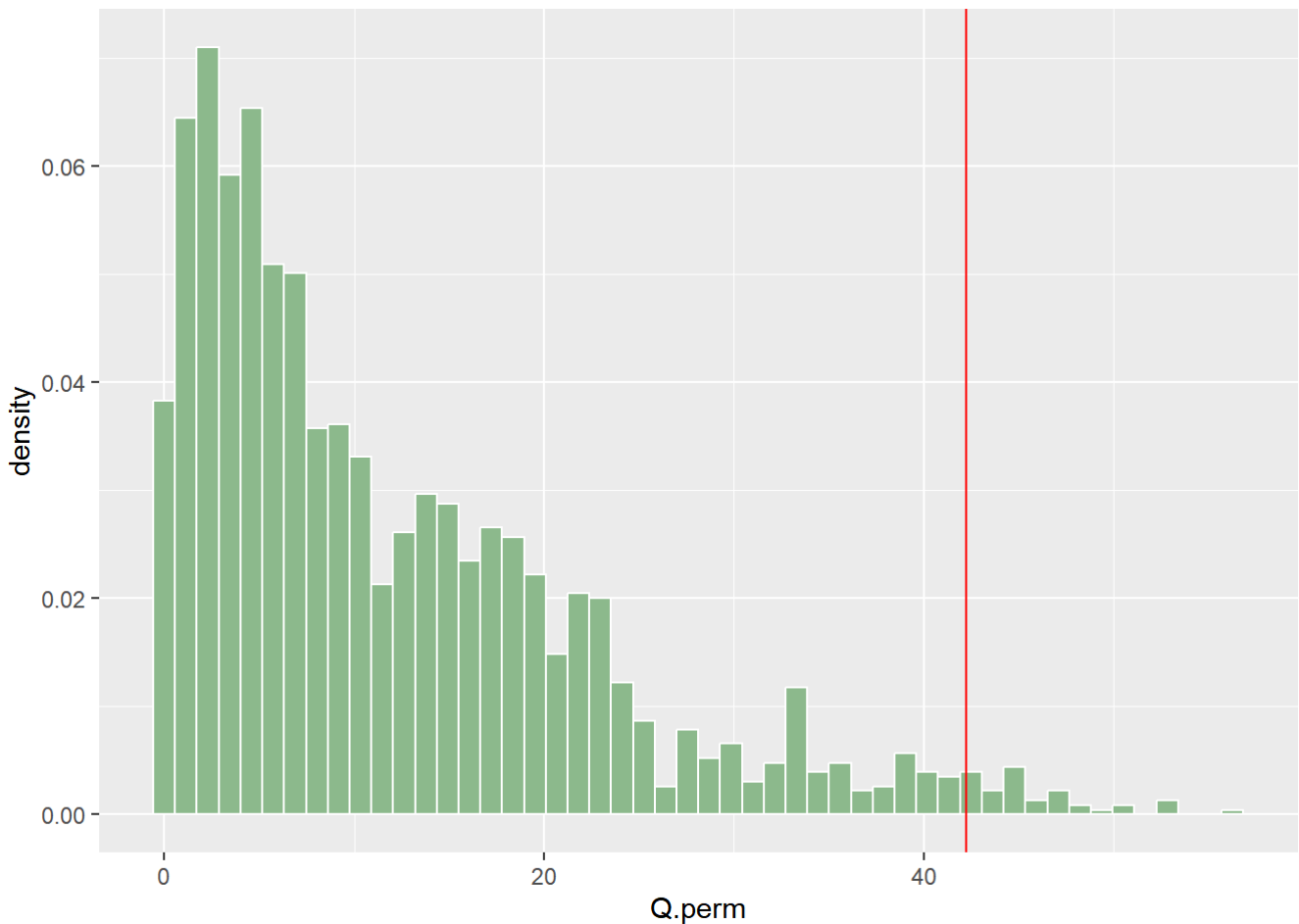
  blocks <- levels(hay$plot)
  treatment.perm <- hay$date
  # block-wise permutation
  for (j in 1:length(blocks)) {
    treatment.perm[hay$plot == blocks[j]] <-
      sample(hay$date[hay$plot == blocks[j]], replace=FALSE)
  }

  Q.perm[r] <- getAlignedQ.CBD(hay$kg, treatment.perm, hay$plot)

}

data.frame(Q.perm=Q.perm) %>% ggplot(aes(x=Q.perm)) +
  geom_histogram(aes(y=..density..), fill= "darkseagreen", color="white", bins=50) +
  geom_vline(xintercept=Q.obs, color='red')

```



```
( p.value <- (sum(Q.perm>=Q.obs)+1)/(Num.perm+1) )
```

[1] 0.0195 < 0,05

(4) Page's test for the ordered alternative: $H_a : t_1 \leq t_2 \leq t_3$

순서가 정해진 H_a 를 검정.

```
DescTools::PageTest(kg ~ date | plot, data=hay) # exact test
```

Page test for ordered alternatives

data: kg and date and plot

L = 79, p-value = 0.02879

cutting date가 늘어날수록
kg가 커지는 경향이 있는가?
Sept.1 ≤ Sept.15 ≤ Sept.30

A normal approximation will be returned for large k & n

```
# rank within a block
```

```
hay <- hay %>% group_by(plot) %>% mutate(rankinblock = rank(kg))
```

```
( ranksum <- aggregate(rankinblock ~ date, data=hay, sum) )
```

date rankinblock

1	Sept.1	9
2	Sept.15	11
3	Sept.30	16

```
( L.obs <- sum(c(1,2,3)*ranksum[,2]) ) # Page statistic
```

[1] 79

$$L = 1R_1 + 2R_2 + 3R_3 = 79$$

이 data가 늘어질수록 R2 커진다면, 3R3 값이 커지고 그에 따라 L도 커짐.

```
# If the alternative hypothesis is true, we expect a large value of L.obs
```

Measures of association 연속형 변수의 관계를 요약하고 검정

The data in "heights.csv" are the mother's heights and daughter's heights (in inches) of 11 mother-daughter pairs.

```
heights <- read.csv("heights.csv")
head(heights)
```

	mother	daughter
1	70	67
2	69	64
3	65	62
4	64	64
5	66	69
6	65	70

(1) Pearson's correlation coefficient

```
x <- heights$mother; y <- heights$daughter
cor(x, y) # pearson
```

[1] 0.4569098

```
cor.test(x, y)
```

Pearson's product-moment correlation

data: x and y

t = 1.541, df = 9, p-value = 0.1577 **70.05** $\leadsto$ not significant.

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.1969458 0.8294436

sample estimates:

cor

0.4569098

```
fit <- lm(y ~ x)
summary(fit)
```

Call:

lm(formula = y ~ x)

Residuals:

Min	1Q	Median	3Q	Max
-5.8745	-2.3033	0.1255	1.9833	5.6485

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	25.0042	26.5097	0.943	0.370
x	0.6192	0.4019	1.541	0.158

Residual standard error: 3.746 on 9 degrees of freedom

Multiple R-squared: 0.2088, Adjusted R-squared: 0.1209

F-statistic: 2.375 on 1 and 9 DF, p-value: 0.1577

$$\hat{\beta}_1 = \frac{\sum y_i}{\sum x_i} = r \sqrt{\frac{\sum y_i^2}{\sum x_i^2}}$$

$\Rightarrow H_0: \beta_1 = 0$ (in simple reg)
 $\approx \Leftrightarrow H_0: \rho = 0$

$$r^2 = (0.4569)^2 \approx 0.2088 = R^2$$

(2) Spearman's rank correlation coefficient

```
cor(x, y, method="spearman")
```

각 변수를 순위로 바꾼 뒤

Pearson 상관계수 계산한 거

```
[1] 0.4282821
```

```
r <- rank(x); s <- rank(y)
(spear.r.obs <- cor(r, s))
```

```
[1] 0.4282821
```

```
cor.test(x, y, method="spearman") # exact test
```

Warning in cor.test.default(x, y, method = "spearman"): Cannot compute exact p-value with ties

따 ties 있으면, 근사값 계산.

Spearman's rank correlation rho

data: x and y

S = 125.78, p-value = 0.1888

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho
0.4282821

$H_0: X$ 와 Y 는 서로 관련이 없다.

```
# permutational test using spearman corr
n <- length(x)
spear.r.perm <- replicate(1999, cor(r, s[sample(1:n, n, replace=FALSE)]))
(p.value <- (sum(spear.r.perm >= spear.r.obs) + 1) / 2000) * 2
```

```
[1] 0.193
```

$(x_2, y_2), (x_5, y_5) \Rightarrow x_2 < x_5 \Rightarrow y_2 < y_5$ 일지
 $\Rightarrow y_5 < y_2$ 아님

(3) Kendall's τ (순서가 일컫는 쌍의 수) - (순서가 어긋나는 쌍의 수)
전체 쌍의 수

```
tau.obs <- cor(x, y, method="kendall")
pnorm(tau.obs, sd=sqrt( (4*n+10)/(9*n*(n-1)) ), lower.tail=FALSE)*2 # large sample test
```

```
[1] 0.1829191
```

```
( p.value <- (1-SuppDists::pKendall(tau.obs, n))*2 ) # exact test
```

```
[1] 0.1645733
```

·X· } Pearson: 직선관계
Spearman: 순위 간 관계
Kendall: 순서가 잘 맞는지 확인

Tests for contingency tables

(1) Chi-square test for independence

다음 시기에... <https://raw.githubusercontent.com/alanagresti/categorical-data/master/Political.dat>은 2450명의 남/여를 대상으로 정치 성향을 조사한 데이터이다.

```
political <- read.table(
  "https://raw.githubusercontent.com/alanagresti/categorical-data/master/Political.dat",
  header = TRUE
)
head(political)
```

```
person gender party
1      1 female  Dem
2      2 female  Dem
3      3 female  Dem
4      4 female  Dem
5      5 female  Dem
6      6 female  Dem
```

```
( gendergap <- xtabs(~gender+party, data=political) )
```

```
      party
gender  Dem Ind Rep
female 495 590 272
male   330 498 265
```

$H_0 : p_{ij} = p_{i.}p_{.j}$ (성별과 정치 성향은 독립이다)를 검정하고자 한다.

```
chisq.test(gendergap)
```

Pearson's Chi-squared test

data: gendergap

X-squared = 12.569, df = 2, p-value = 0.001865

```
chisq.test(gendergap)$expected # counts under the null
```

	party	Dem	Ind	Rep
female	456.949	602.6188	297.4322	
male	368.051	485.3812	239.5678	

(2) Permutational test using chi-square statistic

Consider two treatment methods of relieving pain.

```
painrelief<- data.frame(patient=1:7,
                        satisfaction=c("Not","Not","Somewhat","Somewhat","Somewhat","Very","Very"),
                        treatment=c("P","P","P","P","S","S","S"))
( treatmentgap <- xtabs(~treatment + satisfaction, data=painrelief) )
```

	satisfaction		
treatment	Not	Somewhat	Very
P	2	2	0
S	0	1	2

Note: Row sums are considered fixed.

$H_0 : p_{j|2} = p_{j|2}$ for $j = 1, 2, 3$ 을 검정하고자 한다.

```
( chisq.obs <- chisq.test(treatmentgap)$statistic )
```

Warning in chisq.test(treatmentgap): Chi-squared approximation may be incorrect

X-squared
4.277778

```
# Permute obs. among rows
treatment.perms <- unique(combinat::permn(painrelief$treatment))
( Num.perm <- length(treatment.perms) )
```

[1] 35

```
chisq.perm <- vector(length=Num.perm)
for (i in 1:Num.perm) {

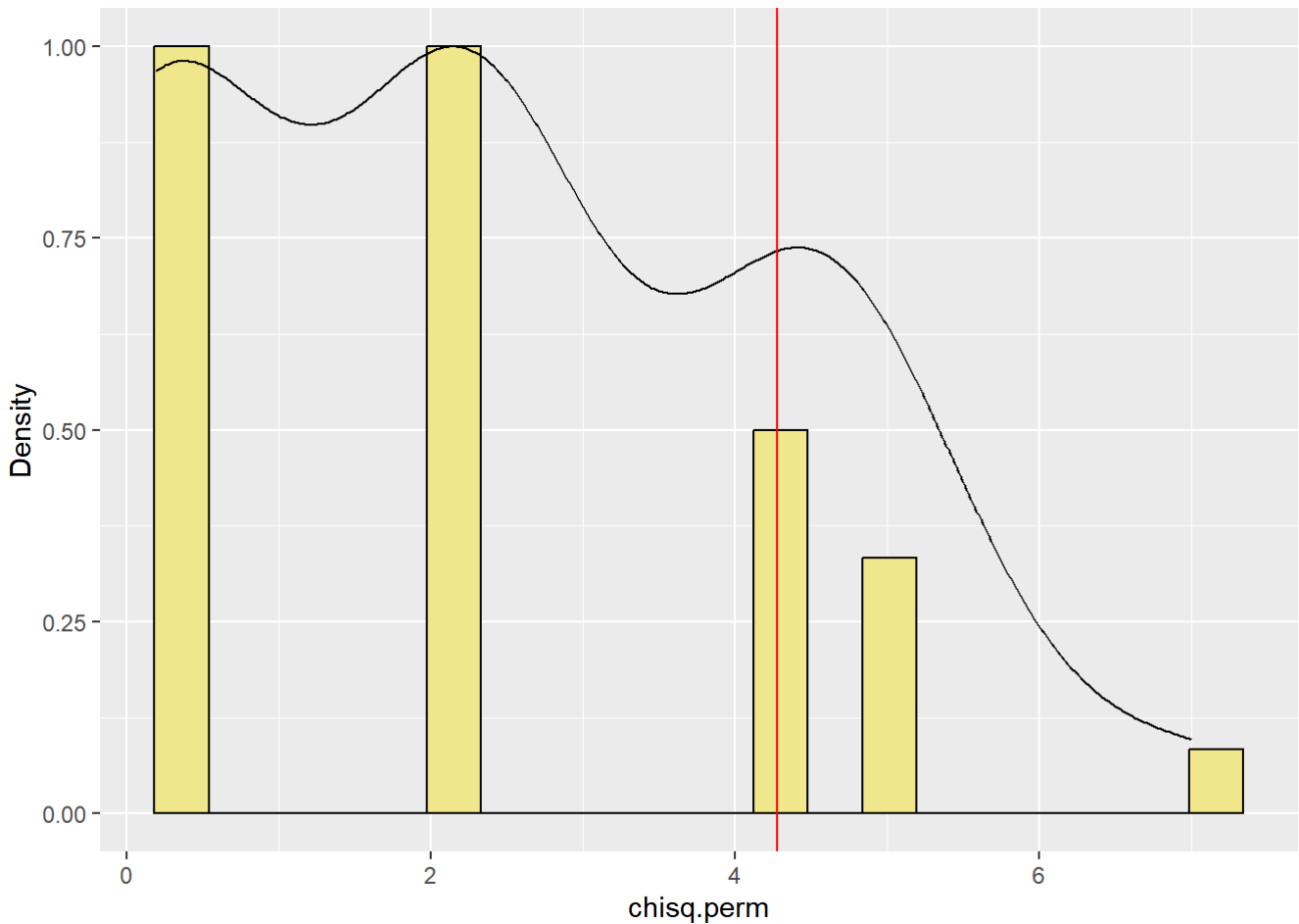
  painrelief.perm <- data.frame(patient=painrelief$patient,
                                satisfaction=painrelief$satisfaction,
                                treatment=treatment.perms[[i]])

  chisq.perm[i] <- chisq.test(xtabs(~treatment + satisfaction,
                                    data=painrelief.perm))$statistic
}

( p.value <- sum(chisq.perm>=chisq.obs)/Num.perm )
```

[1] 0.3142857

```
data.frame(chisq.perm=chisq.perm) %>% ggplot(aes(x=chisq.perm)) +
  geom_histogram(aes(y = ..ncount..), bins = 20, col = 1, fill = "khaki") +
  geom_density(aes(y = ..scaled..)) +
  labs(y = "Density") +
  geom_vline(xintercept=chisq.obs, col = "red")
```



(3) Fisher's exact test for small samples

A colleague of Fisher's claimed that she could distinguish whether milk or tea was added to the cup first when drinking tea.

$H_0 : \theta = 1$ vs $H_a : \theta > 1$

Poured First / Lady says	tea	milk
tea	3	1
milk	1	3

$H_0 : \theta = 1$ 를 검정하고자 한다. (θ 는 odds ratio)

```
tea <- rbind(c(3,1),c(1,3))
theta <- (tea[1,1]*tea[2,2])/(tea[1,2]*tea[2,1])
```

```
fisher.test(tea)$p.value # two-sided
```

→ 여기서 정확한
p-value 계산 가능

```
[1] 0.4857143
```

```
fisher.test(tea, alternative="g")$p.value # one-sided
```

```
[1] 0.2428571
```

(4) McNemar's test

Counts of patients are given in the table below, with the diagnosis (disease: *present* or *absent*) before treatment given in the rows, and the diagnosis after treatment in the columns.

Before / After	present	absent
present	101 (X_{TT})	103 (X_{TB})
absent	77 (X_{BT})	33 (X_{BB})

$H_0 : X_{BT} = X_{TB}$ 를 검정하고자 한다.

```
( disease <- rbind(c(101,103),c(77,33)) )
```

```

      [,1] [,2]
[1,]  101  103
[2,]   77   33

```

```
n <- disease[1,2] + disease[2,1]
1-pbinom(102, size=n, prob=0.5) # exact P-value
```

```
[1] 0.03105338
```

```
1-pnorm((102-0.5*n)/sqrt(0.25*n)) # Normal approx.
```

```
[1] 0.03681914
```

(5) Mantel-Haenszel test for multiple strata

Suppose a study is done at three medical centers to compare Procedures I and II for treating a particular condition (*complications occur (C)* or *no complications occur (NC)*).

Center 1	C	NC	Center 2	C	NC	Center 2	C	NC
I	2	8	I	3	9	I	1	9
II	7	8	II	14	28	II	13	17

$H_0 : \theta(k) = 1$ for $k = 1, 2, 3$ 를 검정하고자 한다.

[illegible]

```
center = c("1", "2", "3"))
```

```
conti.tables
```

```
, , center = 1
```

```
      complication
procedure C NC
  I    2  8
  II   7  8
```

```
, , center = 2
```

```
      complication
procedure C NC
  I    3  9
  II  14 28
```

```
, , center = 3
```

```
      complication
procedure C NC
  I    1  9
  II  13 17
```

```
mantelhaen.test(conti.tables, correct=F) # approximated by a chi-square dist.
```

Mantel-Haenszel chi-squared test without continuity correction

```
data: conti.tables
```

```
Mantel-Haenszel X-squared = 4.5395, df = 1, p-value = 0.03312
```

```
alternative hypothesis: true common odds ratio is not equal to 1
```

```
95 percent confidence interval:
```

```
 0.1297358 0.9414502
```

```
sample estimates:
```

```
common odds ratio
```

```
 0.3494851
```